

Prompt Engineering for Improved Performance



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Course Overview



Harness capabilities of language models

Advanced prompting techniques

Prompt optimization

Reduce latency and improve performance



Module Overview



Prompting at a Glance

- Prompting basics
- Common prompting techniques
 - Zero-shot learning
 - Few-shot learning
 - Instruction prompting
- Prompting failures





Prompting at a Glance

Prompting Basics



What is Prompt Engineering?



Prompt Engineering

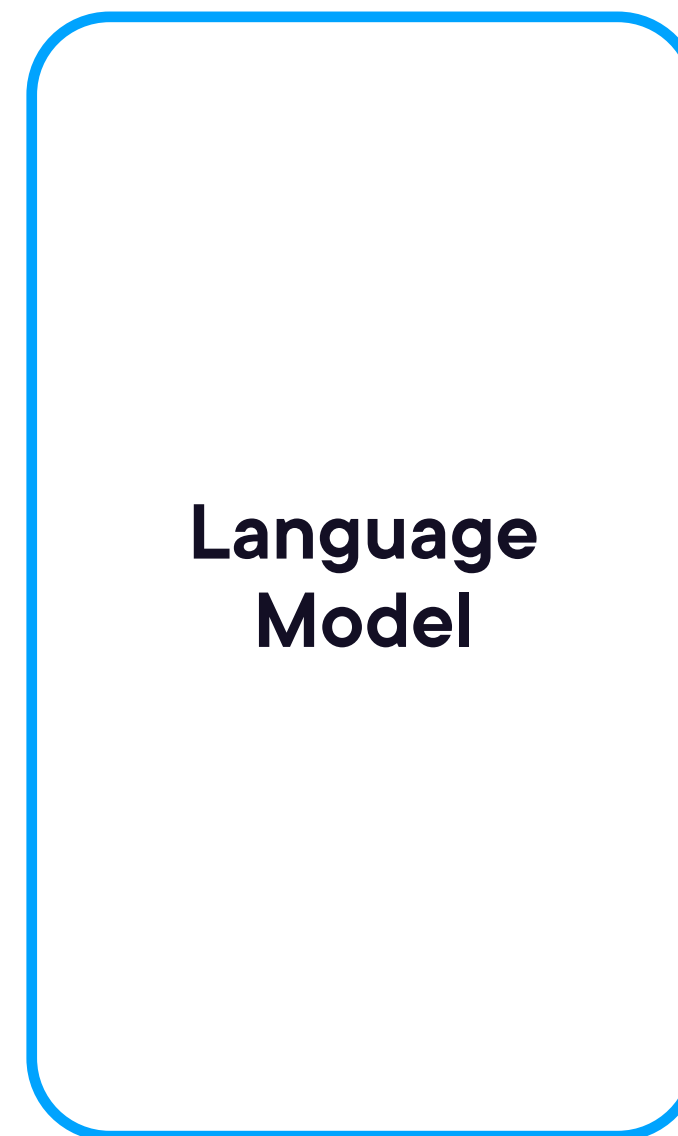
Prompt

A purple rounded rectangle containing five horizontal purple lines, representing a text input field for a prompt.

Input



Language
Model



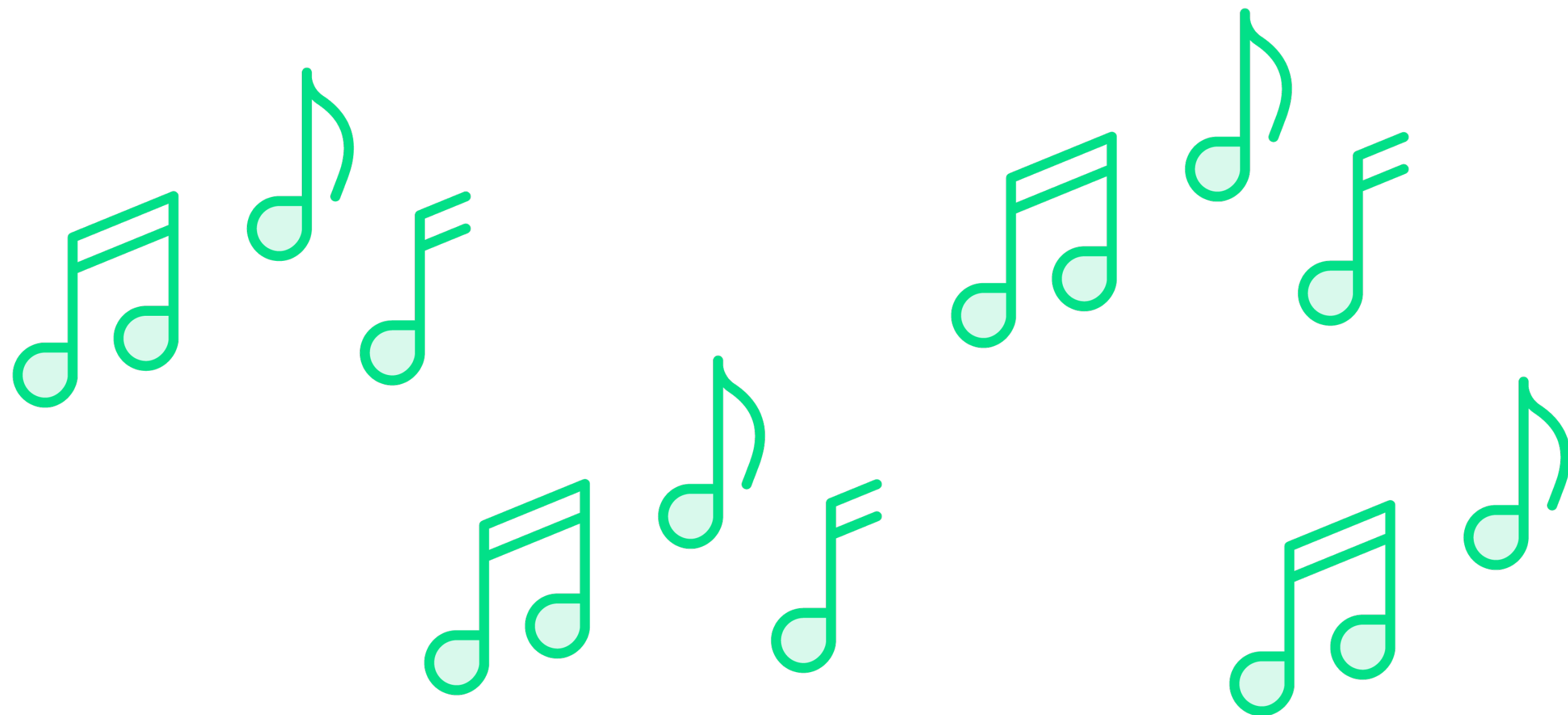
Output



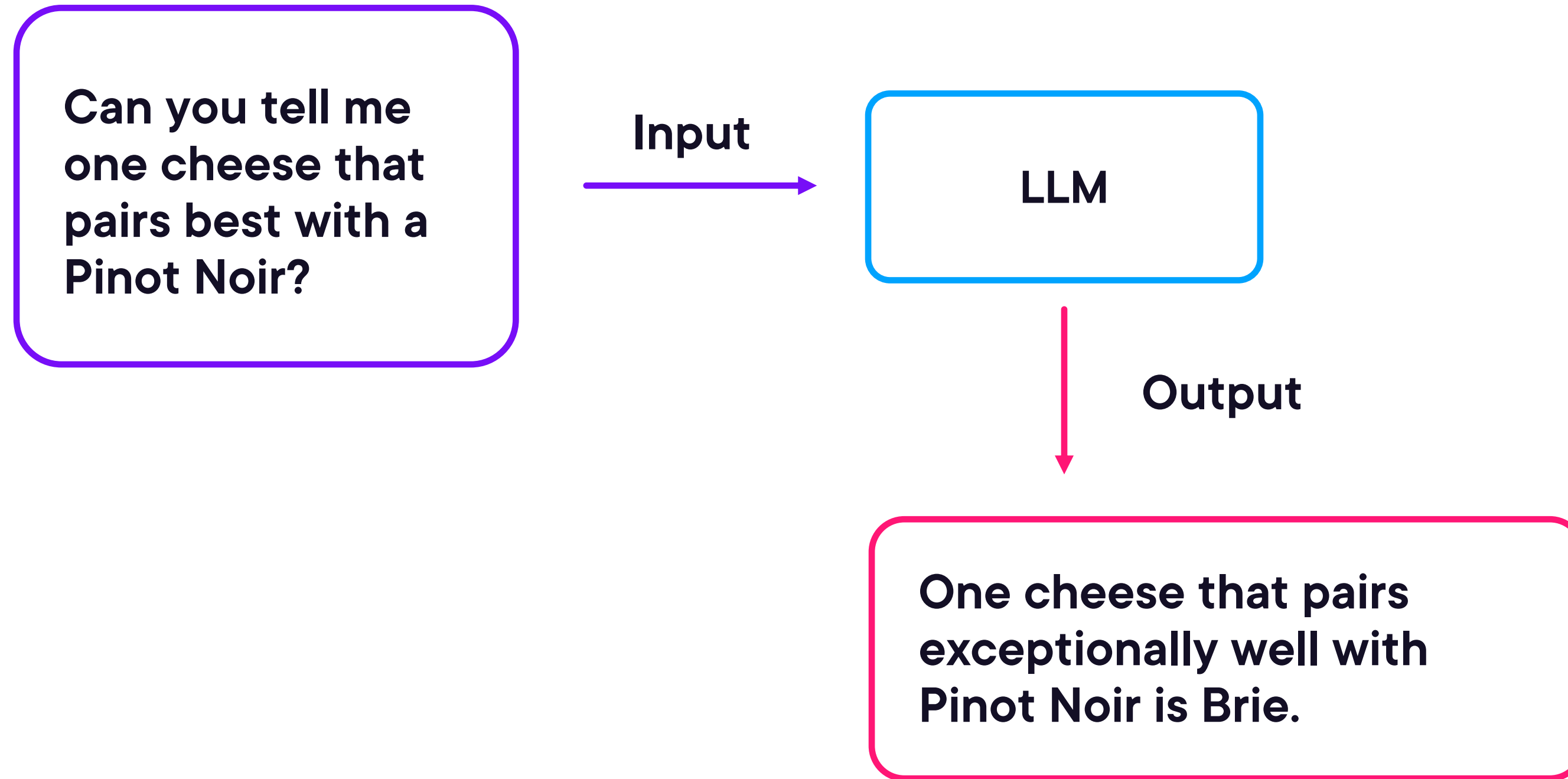
Generated Text

A pink rounded rectangle containing five horizontal pink lines, representing a text output field for generated text.

Prompts Guide Language Models



Simple Prompt





Instruction Prompt

Proofread and correct this article summary. Make it more humorous. Ensure it follows the APA style guide and targets a technical reader. Return your output in markdown format.

Summary: {Text}



Elements of a Prompt

Context

Input Data

Instructions

Examples

Constraints



// Sample Prompt

You are an AI assistant that generates personalized book recommendations based on user preferences.

User profile - Age: 25, Preferred genres: Fantasy, Mystery, Science Fiction, Favorite authors: J.K. Rowling, Agatha Christie

Can you recommend 5 books for this user?

◀ **Context**

◀ **Input Data**

◀ **Instruction**



// Sample Prompt - Continued

Example Input: User profile - Age: 25,
Preferred genres: Fantasy, Mystery,
Science Fiction, Favorite authors: J.K.
Rowling, Agatha Christie.

Example Output: Book recommendation -
"The Night Circus" by Erin Morgenstern.
It combines elements of fantasy and
mystery, making it a captivating read.
After finishing it, I recommend diving
into "The Girl with the Dragon Tattoo"
by Stieg Larsson for an intriguing crime
thriller.

Ensure the generated recommendations are
within the user's preferred genres and
align with their favorite authors. Do
not recommend the books in the example.
Also, maintain a friendly and engaging
tone throughout the generated response.

◀ **Examples**

◀ **Constraints**



Demo



Let's see ChatGPT's response to this Instruction Prompt.





Iterative Nature of Prompt Development



Iterative Process

Which cheese pairs best with a Pinot Noir?

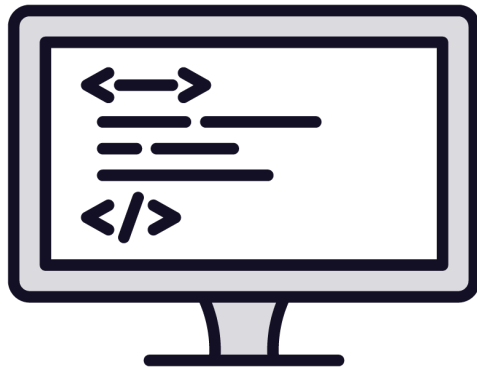


Can you tell me one cheese that pairs best with a Pinot Noir?



Can you tell me one French cheese that pairs best with a Pinot Noir?





Hands-on Challenge

Option 1: Explore ChatGPT

<https://chat.openai.com>

Option 2: Explore OpenAI's Playground

<https://platform.openai.com/playground>

Option 3: Jupyter Notebook

Setup an OpenAI account.



Summary



Prompting Basics

- Helps achieve optimal results
- Contains several elements
 - Context
 - Input data
 - Instructions
 - Examples
 - Constraints



Up Next:

Common Prompting Techniques



Common Prompting Techniques



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Overview



Zero-shot learning

Few-shot learning

Instruction prompting

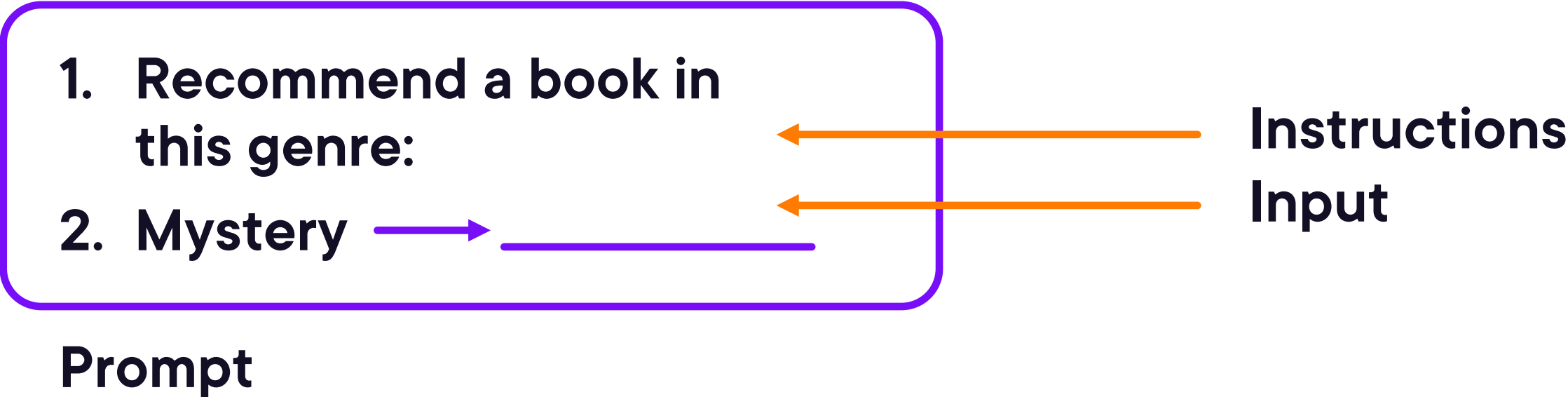




Zero-shot Learning



Example: Zero-shot Learning



Demo



Zero-shot learning

- Chat Completions API
- Authenticate API Key
- OpenAI's Python Library



Improve Model Performance

Context

**Example Input
and Output**

Constraints





Few-shot Learning



Demo



Few-shot learning

- Chat Completions API
- Authenticate API Key
- OpenAI's Python Library





Few-shot learning:

- Large number of tokens
- Charged per token
- Conciseness is crucial





Instruction Prompting



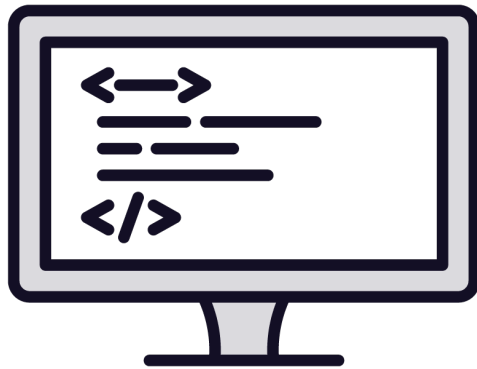
Demo



Instruction Prompting

- Story writing
- Code generation
- RESTful API design





Hands-on Challenge

Jupyter Notebook & OpenAI Python Library

Tips:

- Create an account for the OpenAI API
- Generate an API key for authentication
- Place API key in .env file
- Use OpenAI's Python Library to invoke the Chat Completions API



Summary



Zero-shot learning

Few-shot learning

Instruction prompting



Up Next:

Prompting Failures



Prompting Failures



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Overview



Prompt failures

Mitigation strategies





Why Do Prompts Fail?



Failure Reasons

**Prompts that
are vague**

**Bias and leading
language**

**Evolving language
and knowledge**





Mitigation Strategies



Options

Improve the prompt

**Fine-tune the
foundation model**

**Solve your problem a
different way**



Comparison: By Model Type

Foundation Model: GPT

GPT-3

Broader range of natural language capabilities

Knowledgeable up to 2020

Text-in and text-out

VS

Fine-tuned Model: ChatGPT

GPT 3.5

Fine-tuned for conversational dialogue

Knowledgeable up 2022

Conversation-in and text-out

Web interface on top of GPT 3.5



Summary



Prompt failures

- Prompts that are too vague
- Bias and leading language
- Evolving language and knowledge

Mitigation strategies

- Improve the prompt
- Fine-tune the foundation model
- Solve your problem another way

Module Summary



Prompting at a Glance

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 - Instruction prompting
- Prompting failures



Up Next:

Master Advanced Prompting Techniques Module

